

1. O/N/23/21/Q2

A lorry (truck) is at rest on a straight, horizontal road.

A constant driving force of $5.4 \times 10^4 \text{ N}$ acts on the lorry and it accelerates forwards.

(a) Define what is meant by 'acceleration'.

.....
.....
..... [2]

(b) As the lorry moves forwards, work is done on it by the driving force.

(i) State the equation that defines work done.

Refer to the direction of the force exerted.

.....
..... [1]

(ii) Calculate the work done on the lorry by the driving force when the lorry travels a distance of 280 m.

work done = J [2]

(iii) At one point on the road, the kinetic energy of the lorry is $3.2 \times 10^6 \text{ J}$ and its speed is 12 m/s.

Calculate the mass of the lorry.

mass = kg [3]

[Total: 8]

2. M/J/23/22/Q2

Fig. 2.2 shows a rider on an electric scooter.



Fig. 2.2

The scooter contains a battery and a motor to drive the back wheel.

- (a) (i)** State the name of the energy store in the battery.

..... [1]

- (ii)** Describe, in terms of work done, the stages of energy transfer from the energy store in the battery to the kinetic energy of the scooter.

.....
.....
.....
..... [2]

- (b)** The total mass of the scooter and the rider is 70 kg.

Calculate the total kinetic energy of the rider and scooter when the scooter has a speed of 4.0 m/s.

kinetic energy = J [2]

(c) The battery is marked 'energy capacity 0.35 kilowatt-hour (kW h)'.

(i) Define what is meant by a kilowatt-hour.

.....
..... [1]

(ii) The scooter stops working because the battery is totally discharged (flat). This means that there is no more energy stored in the battery.

The battery is then recharged using a 70W power supply.

Calculate the time taken to fully recharge the battery.

time = hours [2]

[Total: 8]

3. O/N/22/21/Q2

A child's toy consists of a flexible track and a model car.

Fig. 2.1 shows a diagram of the toy.

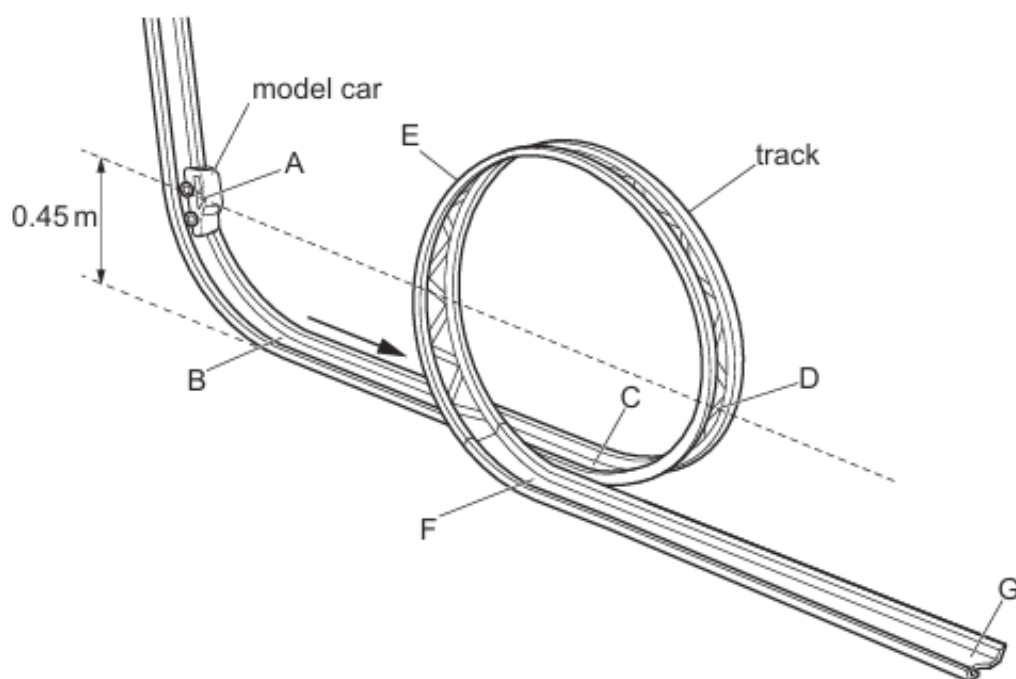


Fig. 2.1

The child first holds the car stationary at point A which is 0.45 m above the horizontal sections of track BC and FG. The mass of the car is 0.12 kg.

The child then releases the car which travels towards point B. Both air resistance and friction between the car and the track are negligible.

The gravitational field strength g is 10 N/kg.

- (a) Calculate the change in gravitational potential energy (g.p.e.) of the car as it travels from A to B.

change in g.p.e. = [2]

(b) Calculate the speed of the car when it reaches B.

speed = [3]

(c) After releasing it, the child expects the car to follow the track along the route ABCDEFG. In fact, the model car does not reach F.

(i) Explain, in terms of energy, why the car does not go past D, which is also 0.45 m above the horizontal track.

.....
..... [1]

(ii) Immediately after being released at A, the car travels to B, to C and then to D.

Describe the motion of the car after it reaches D.

.....
..... [1]

[Total: 7]

4. O/N/22/22/Q2

The foundations that support a building are long concrete cylinders that are pointed at one end. A pile-driver is a machine that forces the pointed concrete cylinders into the ground.

Fig. 2.1 shows a pile-driver.

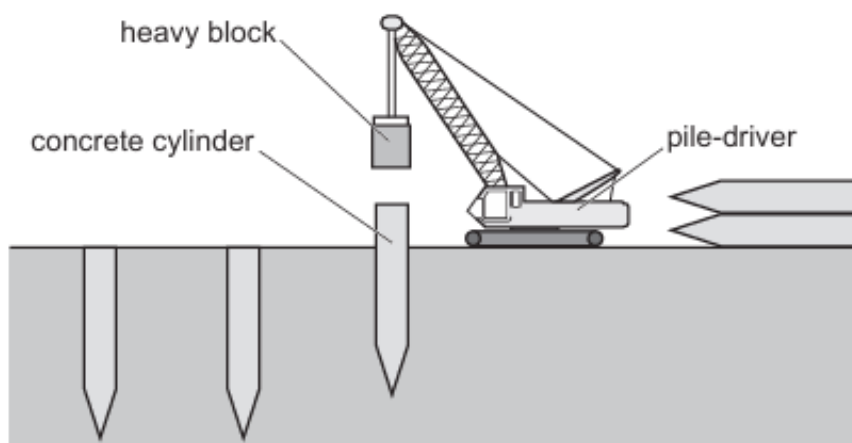


Fig. 2.1

A heavy block of mass $2.9 \times 10^4 \text{ kg}$ is lifted into the air then dropped onto the top of a concrete cylinder. This forces the cylinder into the ground.

(a) Fig. 2.2 shows the heavy block.

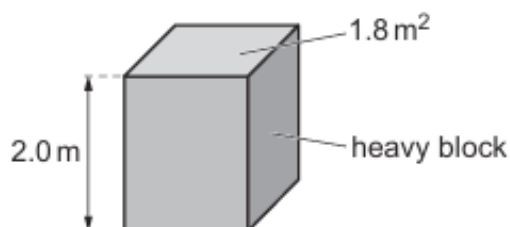


Fig. 2.2

The block is 2.0 m tall and has a cross-sectional area of 1.8 m^2 .

Calculate the density of the material used to make the block.

density = [2]

- (b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.
The gravitational field strength g is 10 N/kg.

(i) Calculate the gravitational potential energy gained by the block.

gravitational potential energy = [2]

(ii) The block is then dropped from rest onto the top of the concrete cylinder.

Calculate the speed of the block just before it hits the concrete cylinder.

speed = [3]

[Total: 7]

5. M/J/22/21/Q2

Fig. 2.1 shows a wind turbine.

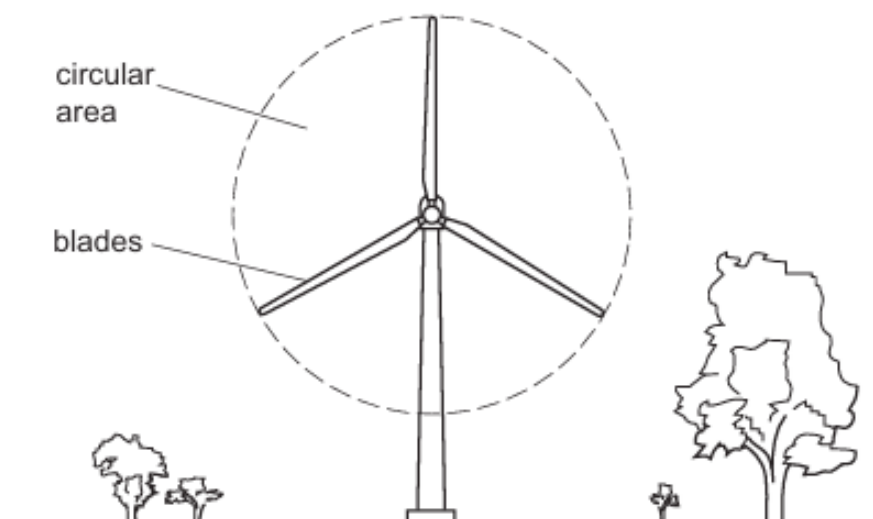


Fig. 2.1

- (a) The wind blows directly towards the turbine with a speed of 12 m/s .

In one second, $60\,000 \text{ kg}$ of air passes through the circular area swept out by the blades.

Calculate the kinetic energy of this mass of air.

kinetic energy = [3]

- (b) A wind turbine releases no carbon dioxide into the atmosphere while generating electricity. Compared with a coal-fired power station, this is an advantage of using a wind turbine.

- (i) State **one** reason why it is important to reduce the amount of carbon dioxide produced.

.....
..... [1]

- (ii) Suggest **one** advantage of using a coal-fired power station compared with a wind turbine.

.....
..... [1]

- (c) A coal-fired power station releases 0.96 kg of carbon dioxide when it generates 1.0 kWh of electrical energy.

(i) Define the kilowatt-hour (kWh).

.....

..... [1]

- (ii) Calculate the mass of carbon dioxide saved when the wind turbine has a power output of 2000 kW and operates for 12 hours.

mass = [1]

- (d) Wind energy is a form of renewable energy.

State the name of **one** other form of renewable energy.

..... [1]

[Total: 8]

6. O/N/21/21/Q4

In a coal-fired power station, coal is burnt and electrical energy is produced.

- (a) State the form of energy in the coal that is transferred to electrical energy.

..... [1]

- (b) (i) State the principle of the conservation of energy.

.....
.....
.....
..... [2]

- (ii) The quantity of electrical energy produced by the power station is much less than the quantity of energy in the coal that is burnt to generate the electricity.

Explain why this is so.

.....
.....
.....
..... [2]

- (c) State **one** environmental consequence of generating electricity in coal-fired power stations.

.....
..... [1]

[Total: 6]

7. M/J/21/22/Q9

(a) Fig. 9.1 represents the conversion of energy in a mobile phone.



Fig. 9.1

(i) State the form of energy stored in the battery.

..... [1]

(ii) State the useful form of output energy from the battery.

..... [1]

(iii) State **two** useful forms of output energy from the mobile phone.

..... [2]

(b) When the battery is in use, the average current is 1.3 mA.

Determine the charge that passes through the battery in a time of 2.0 minutes.

charge = [2]

- (c) A student estimates the thermal energy produced in the battery when the mobile phone is used and uses this value to estimate the efficiency of the battery.

He measures a temperature rise of 5.0°C within the battery in a period of time when the useful energy output from the battery is 5200 J .

The mass of the battery is 110 g and its specific heat capacity is $830\text{ J/(kg }^{\circ}\text{C)}$.

- (i) Calculate the thermal energy (heat) needed to raise the temperature of the battery by 5.0°C .

thermal energy = [3]

- (ii) State what is meant by *efficiency*.

.....
..... [2]

- (iii) Calculate the efficiency of the battery.

efficiency = [2]

- (iv) Suggest and explain **one** reason why the value for the efficiency calculated in (iii) is larger than the actual efficiency of the battery.

.....
.....
.....
..... [2]

[Total: 15]

8. M/J/21/21/Q8(c)

Fig. 8.1 shows a stationary horse and its rider, about to jump over two fences.

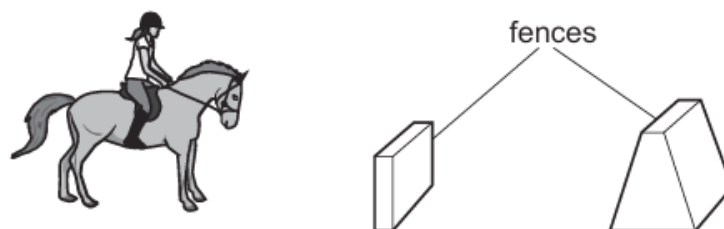


Fig. 8.1

(a) Fig. 8.2 shows a side view of the horse.

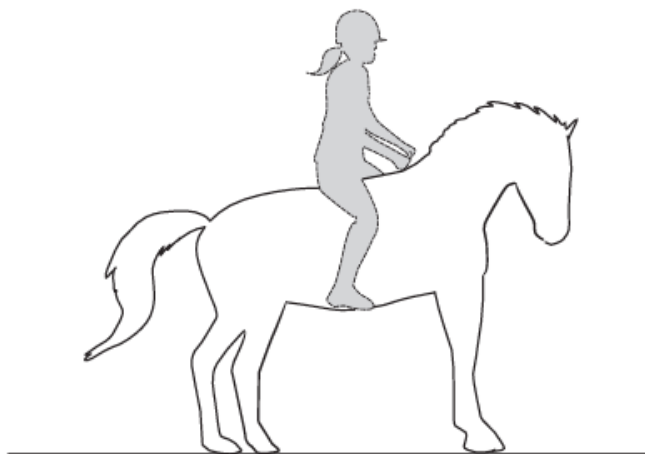


Fig. 8.2

(c) The total mass of the horse and rider is 520 kg.

(i) As they approach a fence, the horse and rider have a total kinetic energy of 4000 J.

Calculate their speed.

speed = [3]

- (ii) The centre of mass of the horse and rider is 1.4 m above the ground.

The maximum potential energy gained by the horse and rider as they jump over the fence is 3000 J.

Calculate the maximum height above the ground of the centre of mass during the jump.

The gravitational field strength $g = 10 \text{ N/kg}$.

height = [3]

9. O/N/20/22/Q2

The total mass of a wheelbarrow and its load is 90 kg. A worker pushes the wheelbarrow and load up a plank of length 2.0 m on to a platform, as shown in Fig. 2.1. The platform is at a height of 0.60 m above the ground.

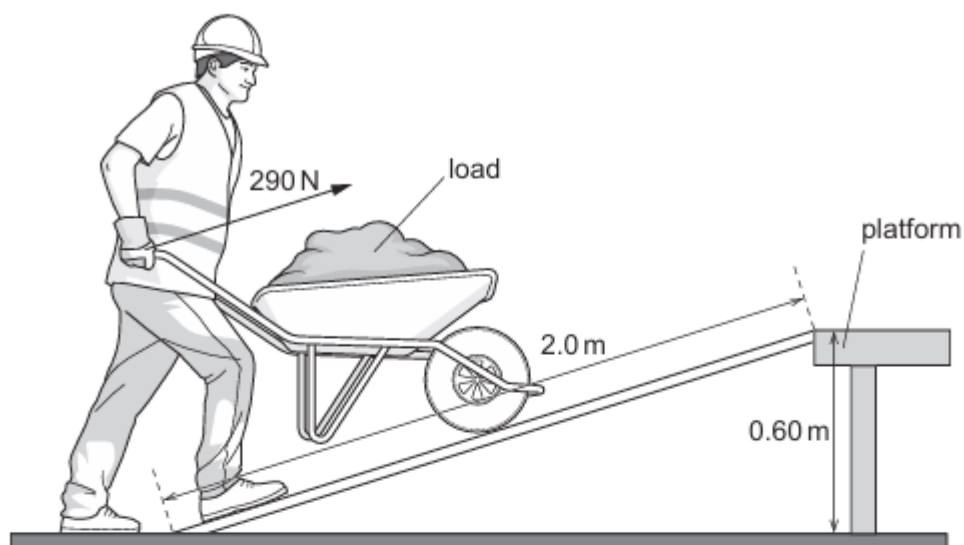


Fig. 2.1

The worker exerts a force of 290 N on the wheelbarrow in the direction in which the wheelbarrow moves.

(a) The gravitational field strength g is equal to 10 N/kg.

(i) Calculate the gravitational potential energy gained by the wheelbarrow and its load.

energy = [2]

(ii) The worker pushes the wheelbarrow 2.0 m along the plank.

Calculate the work done on the wheelbarrow by the worker.

work done = [2]

(iii) Suggest one reason why the answer to **(a)(ii)** is not equal to the answer to **(a)(i)**.

.....
..... [1]

(b) The worker finds that walking up the plank pushing the wheelbarrow is even more inefficient than the answers in (a) suggest.

(i) State what is meant by *efficiency*.

.....

..... [1]

(ii) Suggest **one** reason why this method of lifting the load onto the platform is so inefficient.

.....

..... [1]

[Total: 7]

10. M/J/20/22/Q6

The clothes iron shown in Fig. 6.1 is connected to the electrical mains.

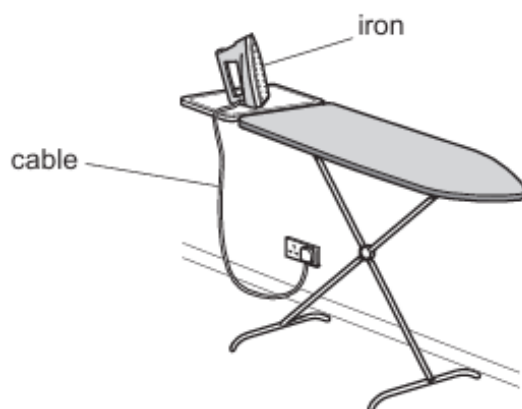


Fig. 6.1

- (a) The boxes in the left column below contain some electrical hazards. The boxes in the right column contain methods of protection from these hazards.

For each hazard, draw **one** line to the appropriate method of protection.

electrical hazard	method of protection
worn insulation on the cable to the iron	earth wire and fuse in plug correctly connected to iron
loose live wire in the iron touches its metal case	circuit breaker correctly connected in circuit
cable becomes too hot because current is too high	visual check of cable before connecting to mains

[2]

- (b) The power of the iron is 1200 W. The cost of 1 kWh of electrical energy is 20 cents (20 c).

- (i) Define the *kilowatt-hour* (kWh).

.....
 [1]

- (ii) The iron is on at full power for 20 minutes.

Calculate the cost of running the iron for this time.

cost = [2]

11. M/J/20/21/Q10

Fig. 10.1 shows a motor lifting a mass. Fig. 10.2 shows part of the circuit diagram of the connections to the motor.

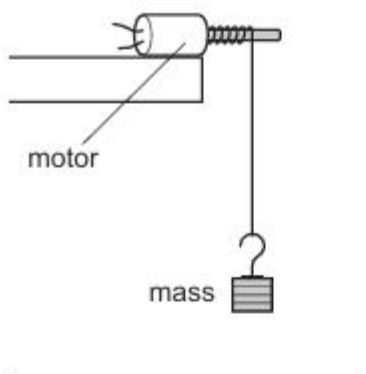


Fig. 10.1

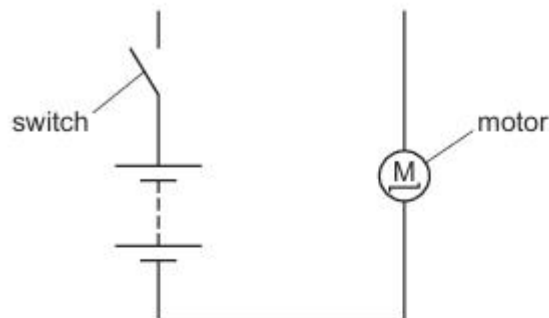


Fig. 10.2

- (a) The current in the motor is 1.5A and the voltage supplied by the battery is 8.0V.
- (i) Complete the circuit diagram in Fig. 10.2 to show an ammeter and a voltmeter in the correct positions to take these measurements while the motor is working. [2]
- (ii) The motor takes 4.0 s to lift the mass.

Calculate the electrical energy transferred to the motor in this time.

energy = [2]

- (iii) The motor lifts the 150 g mass through a height of 80 cm in the 4.0 s.

Calculate the gravitational potential energy gained by the mass.

The gravitational field strength $g = 10 \text{ N/kg}$.

gravitational potential energy = [3]

- (iv) State **two** reasons why the gravitational potential energy gained by the mass is less than the electrical energy supplied to the motor.

1.

2.

[2]

12. O/N/19/21/Q2

A small coin of mass m is initially at rest. It is dropped from a height h above ground level. The coin has a speed v as it hits the ground.

The gravitational field strength g is equal to 10 N/kg .

(a) State an expression for:

- the gravitational potential energy of the coin at a height h above ground level

gravitational potential energy =

- the kinetic energy of the coin when travelling at a speed v .

kinetic energy = [1]

(b) The coin is dropped from the top of a building of height 380 m .

Determine the speed of the coin as it hits the ground. You may ignore air resistance.

speed = [3]

(c) When air resistance is negligible, a heavier coin hits the ground at the same speed as a lighter coin when they are both dropped from the same height.

(i) Explain why.

.....
.....
..... [1]

(ii) When air resistance acts, coins of different masses do **not** hit the ground at the same speed when they are dropped from the same height.

Explain why.

.....
.....
..... [2]

13. M/J/19/21/Q3

A large spring is stretched by an athlete to increase the strength of his arms.

The athlete extends the spring, as shown in Fig. 3.1, and then releases it gently. He extends and releases the spring several times during a period of 60 s.



Fig. 3.1

- (a) State the form of energy stored in the stretched spring and the form of energy stored in the muscles of the athlete.

form of energy in spring

form of energy in muscles

[2]

- (b) Suggest what happens if he extends the spring beyond the limit of proportionality.

.....

..... [1]

- (c) During one extension of the spring, its length increases from 70 cm to 93 cm. The average force exerted by the athlete is 400 N.

- (i) Calculate the work done by the athlete in one extension.

work done = [2]

- (ii) Describe how the athlete can use the value obtained in (i) to determine his average power during the 60 s.

.....

.....

.....

..... [2]

14. M/J/18/22/Q9

Wind energy is a renewable energy source. A wind turbine and generator convert energy in the wind to electrical energy in a generator. Fig. 9.1 shows some wind turbines.

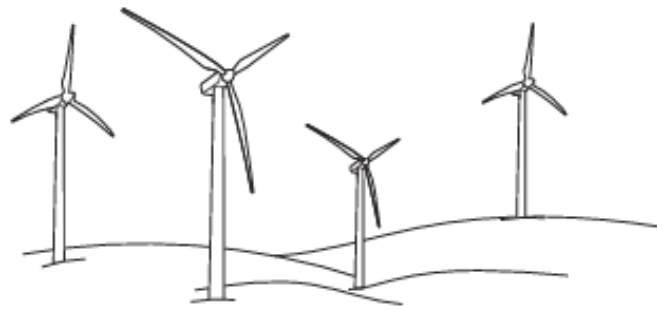


Fig. 9.1

- (a) (i) State what is meant by a *renewable energy source*.

.....
.....
.....[1]

- (ii) State the name of **one other** renewable energy source.

.....
.....[1]

- (iii) Some methods of generating electricity cause much more global warming than wind energy.

1. State **one** method of generating electricity that causes a large amount of global warming.

.....[1]

2. Describe, in outline, how the method you have chosen causes a significant amount of global warming.

.....
.....
.....[1]

- (b)** During a 30 minute period, a mass of 4.2×10^7 kg of air enters the turbine with a speed of 15 m/s.

- (i)** Calculate the kinetic energy of the air that enters the turbine in 30 minutes.

energy =[2]

- (ii)** The electrical power output of the generator connected to the turbine is 8.4×10^5 W.

Calculate the electrical energy output from the turbine in 30 minutes.

energy output =[2]

- (iii)** Calculate the efficiency of the turbine and generator in converting the kinetic energy of the air to electrical energy.

efficiency =[2]

15. M/J/18/21/Q4

A woman does work as she raises a flag by pulling downwards on a rope. This is shown in Fig. 4.1.

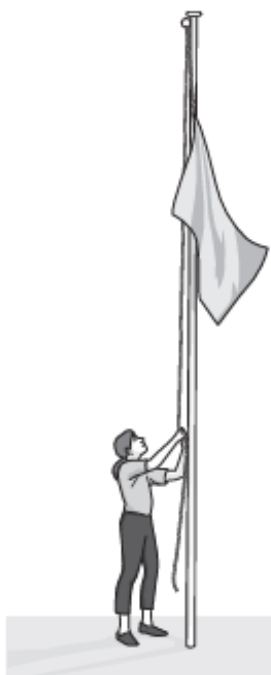


Fig. 4.1

- (a) Define what is meant by *work done*.

.....
.....[2]

- (b) The mass of the flag is 0.35 kg and the flag is raised through a height of 6.0 m.

The gravitational field strength g is 10 N/kg.

Calculate the increase in gravitational potential energy of the flag.

gravitational potential energy =[2]

- (c) Suggest and explain a reason why the work done on the rope is larger than the value calculated in (b).

.....
.....
.....
.....[2]

16. M/J/17/22/Q2

A swing is made by tying rope loosely to the branch of a tree, as shown in Fig. 2.1.

A child swings backwards and forwards several times, starting at the highest point A.

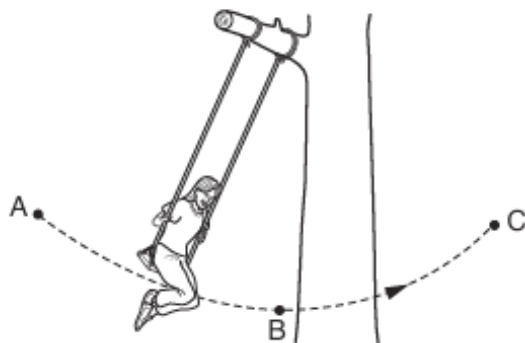


Fig. 2.1

- (a) Explain how another child can obtain an accurate measurement of the time for one complete swing.

.....
.....
.....
.....
.....[2]

- (b) When the child moves from A to B, she falls a vertical distance of 0.60 m. She loses 240 J of gravitational potential energy. The gravitational field strength g is 10 N/kg.

- (i) Calculate the mass of the child.

mass =[2]

- (ii) Suggest two reasons why her kinetic energy at B is not equal to 240 J.

1.
.....
2.
.....

[2]

17. M/J/17/21/Q9

Different energy sources are used to generate electricity.

(a) Energy sources are renewable or non-renewable.

(i) Nuclear energy is described as a non-renewable source.

Explain what is meant by a *non-renewable* energy source.

.....
[1]

(ii) Four of the energy sources used are:

hydroelectric

oil

geothermal

wind

Write the name of these energy sources in the correct column of the table below.

non-renewable	renewable and caused by energy from the Sun	renewable and not caused by energy from the Sun

[3]

(iii) State one way in which using nuclear energy is better for the environment than using oil.

.....
[1]

(iv) State one way in which using nuclear energy is worse for the environment than using oil.

.....
[1]

(b) Fig. 9.1 is a block diagram of a power station that produces electrical energy from oil.

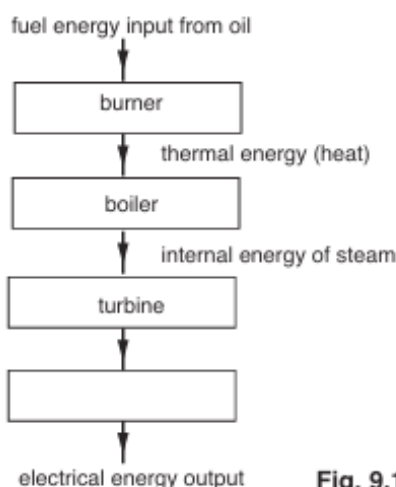


Fig. 9.1

(i) Write the name of the missing part of the power station in the empty box on Fig. 9.1. [1]

(ii) State the form of energy that the turbine possesses.

.....[1]

18. O/N/16/21/Q3

A large power station burns oil to heat water in a boiler.

- (a) State the name of the form of energy stored in the oil before it is burnt.

.....[1]

- (b) (i) State and explain whether oil is a renewable or a non-renewable energy source.

.....
.....
.....[1]

- (ii) Describe one environmental issue caused by burning oil to generate electricity in a power station.

.....
.....[1]

- (c) (i) State what is meant by *efficiency*.

.....
.....[1]

- (ii) The efficiency of the power station is 0.38 (38%). Its electrical power output is 1.9×10^9 W.

Calculate

1. the input power to the power station,

input power =[2]

2. the energy wasted in 2.0 hours.

energy wasted =[2]

19. M/J/16/22/Q2

Fig. 2.1 shows an electric motor lifting a load.

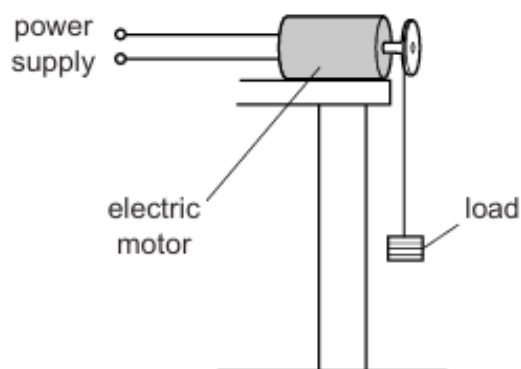


Fig. 2.1

The load, of weight 5.0 N , is raised through a vertical height of 3.5 m at a constant speed. The efficiency of the electric motor is 0.65 (65%).

- (a)** Calculate the increase in gravitational potential energy of the load.

gravitational potential energy =[2]

- (b) (i)** State the formula that relates efficiency, energy input and useful energy output.

.....
.....[1]

- (ii)** Calculate the energy input to the motor.

energy input =[2]

- (c)** Suggest one reason why the efficiency of the motor is less than 1.0 (100%).

.....
.....[1]

20. O/N/15/22/Q8

A spotlight in a television studio is at full brightness. The lamp inside the spotlight is powered by a 230 V supply and the current in the filament of the lamp is 27 A.

- (a) Calculate the power of the lamp.

power = [2]

- (b) The spotlight is kept switched on at full brightness.

- (i) Calculate the energy transformed by the lamp in 30 minutes.

energy = [1]

- (ii) The cost of using one kilowatt-hour (kWh) of electricity is 23 cents.

Calculate the cost of using the spotlight for 30 minutes.

cost = [2]

21. O/N/15/22/Q1

Water is transported to a village in a tank pulled by a tractor.

Fig. 1.1 shows the tank being pulled by a tractor.

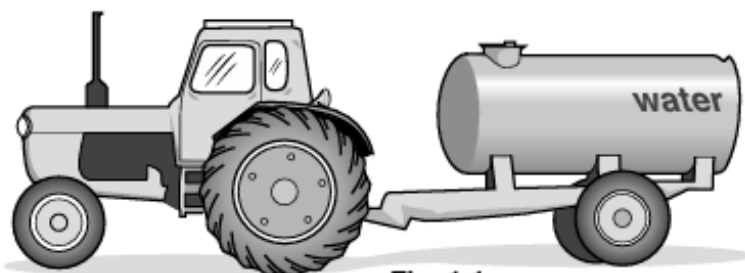


Fig. 1.1

The combined mass of the tractor and the tank is 4100 kg when the tank is empty and 6500 kg when the tank is full of water.

- (a)** The density of water is 1000 kg/m^3 .

Calculate the volume of water in the tank when it is full.

volume = [2]

- (b)** At the start of the journey, the tractor and tank accelerate from rest along a straight, horizontal road. As their speed increases, one form of energy is decreasing.

- (i)** State the name of the form of energy that is decreasing.

..... [1]

- (ii)** Explain what happens to this energy.

.....
.....
.....
..... [2]

- (c)** The village is located on a mountain at a vertical height of 850 m above the water supply. The gravitational field strength g is 10 N/kg .

Calculate the gravitational potential energy gained by the water as it is transported from the supply to the village.

energy gained = [2]

22. O/N/15/21/Q3

A lorry of mass 4.4×10^4 kg travels along a straight, horizontal road at 20 m/s.

- (a) Calculate the kinetic energy of the lorry.

kinetic energy = [3]

- (b) The lorry driver sees an obstruction ahead and applies the brakes. The lorry slows down and stops. The lorry stops a distance of 40 m from where the driver first applies the brakes.

As it slows down, the kinetic energy of the lorry decreases to zero. The kinetic energy decreases as work is done by the braking force exerted on the lorry.

- (i) State the formula that relates the work done to the braking force.

.....
.....
..... [1]

- (ii) Calculate the braking force exerted on the lorry.

force = [2]

23. M/J/13/21/Q9

Fig. 9.1 shows a conveyor belt carrying suitcases into an aeroplane.

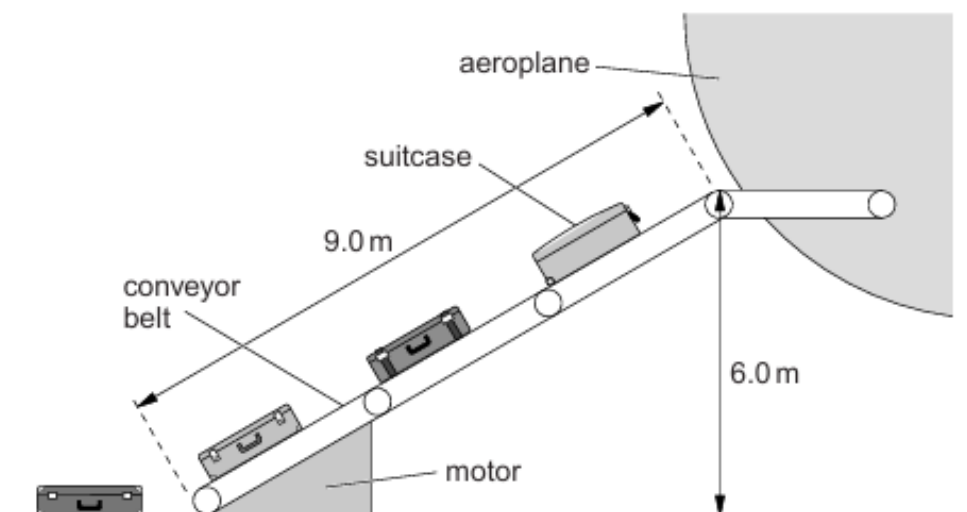


Fig. 9.1 (not to scale)

An electric motor drives the conveyor belt.

(a) A suitcase of mass 20 kg is lifted from the ground into the aeroplane.

(i) Explain what is meant by *mass* and by *weight*.

.....

 [2]

(ii) The gravitational field strength g is 10 N/kg.
 Calculate the increase in the gravitational potential energy of the suitcase.

increase in potential energy = [2]

(iii) The suitcase takes 12 s to travel 9.0 m along the conveyor belt.
 Calculate the kinetic energy of the suitcase.

kinetic energy = [3]

- (iv) The electric motor is powered by an input voltage of 230V and a current of 1.5A. Calculate the electrical energy input to the motor in 12 s.

energy =[2]

- (v) 1. State the principle of the conservation of energy.

.....
.....
.....[2]

2. Explain how this principle applies to the lifting of the suitcases into the aeroplane.

.....
.....
.....[1]

- (b) Some of the electrical energy for the motor comes from renewable energy sources.

- (i) Explain what is meant by a *renewable energy source*.

.....
.....
.....[1]

- (ii) State two renewable energy sources.

1.
2.
[2]

24. O/N/12/21/Q3

Fig. 3.1 shows an electric motor being used on a building site to lift bricks.

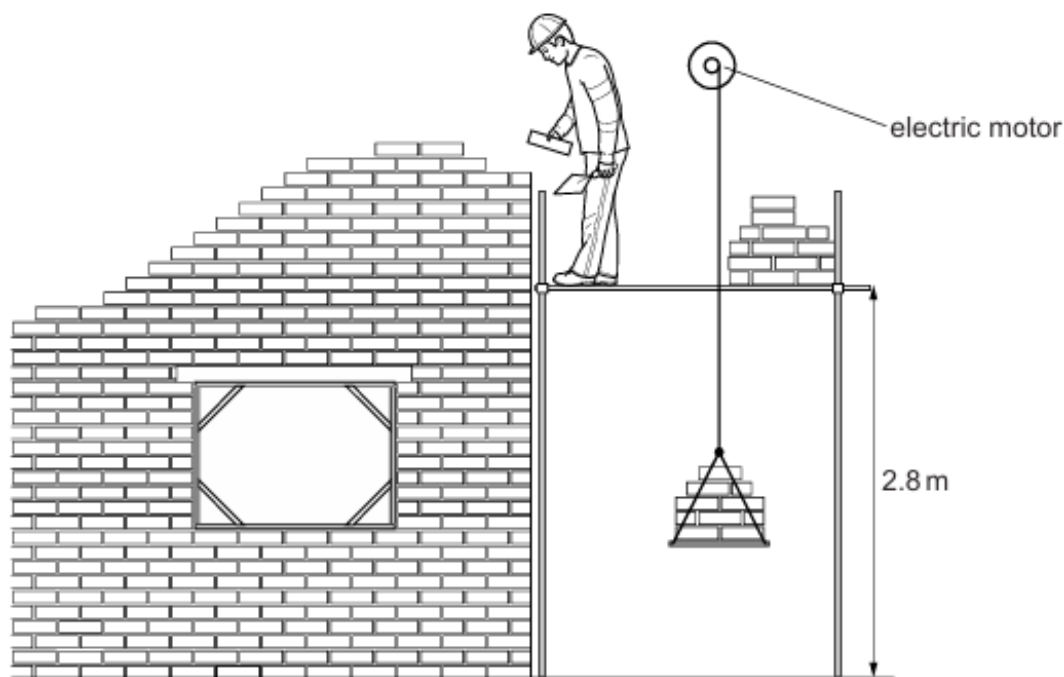


Fig. 3.1

The gravitational field strength is 10 N/kg . The motor lifts bricks of total mass 54 kg through a height of 2.8 m in 3.0 s .

(a) Calculate

(i) the work done in lifting the bricks,

work done = [2]

(ii) the useful output power of the motor.

power = [2]

(b) Suggest two reasons why the electrical power supplied to the motor is greater than its useful output power.

1.

2.

[2]

25. M/J/15/22/Q3

Two different kettles are used to heat water, as shown in Fig. 3.1.

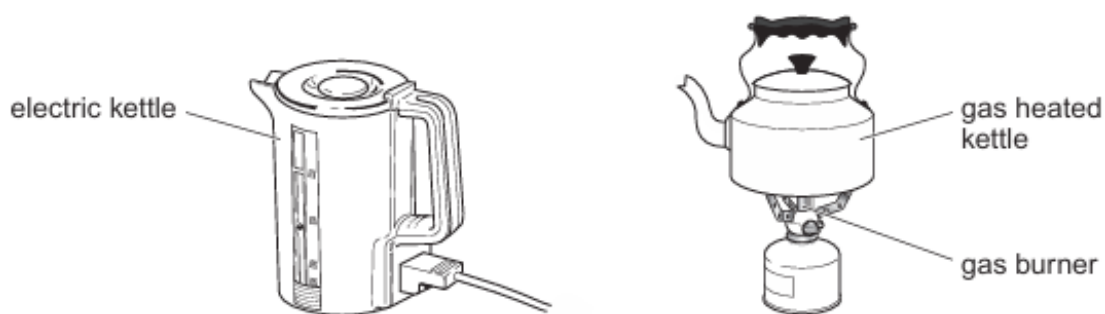


Fig. 3.1

Data for the two kettles is shown in Fig. 3.2.

	energy supplied to the kettle in one minute / J	thermal energy (heat) supplied by the kettle to the water in one minute / J
electric kettle	120 000	95 000
gas heated kettle	130 000	90 000

Fig. 3.2

(a) (i) Calculate the efficiency of the electric kettle.

efficiency = [2]

(ii) Calculate the useful power of the gas heated kettle.

power = [2]

26. O/N/14/21/Q9

Fig. 9.1 shows the dam and reservoir of a hydroelectric power station.

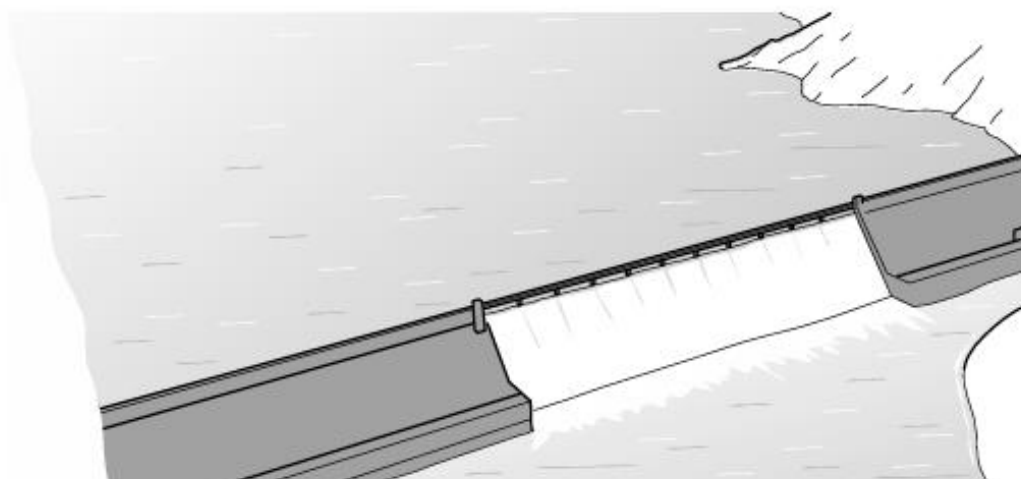


Fig. 9.1

A hydroelectric power station uses a renewable energy source.

(a) State two other renewable energy sources.

1.

2.

[2]

(b) When the power station operates at full capacity, the electrical power output is $6.8 \times 10^9 \text{ W}$.

(i) 1. Calculate the electrical energy output of the power station if it operates at full capacity for one year.

energy =[1]

2. Suggest why, in practice, the power station does not operate at full capacity throughout the year.

.....

.....[1]

- (ii) The water surface in the reservoir that supplies the hydroelectric power station is at a vertical height of 170m above the turbines. In one hour, 1.6×10^{10} kg of water flows from the reservoir through the turbines. The gravitational field strength g is 10 N/kg.

1. Calculate the gravitational potential energy that is converted into other forms of energy in one hour.

energy =[2]

2. Calculate the efficiency of the power station when operating at full capacity.

efficiency =[2]

3. Suggest two reasons why the efficiency of the power station is less than 1.0 (100%).

1.

2.

[2]

27. O/N/12/22/Q3

A man of mass 75 kg falls from a platform high above a lake.

Fig. 3.1a shows the man tied to the platform by a long elastic rope (bungee).

Fig. 3.1b shows the man when he has fallen 20 m. After this point the rope begins to stretch.

Fig. 3.1c shows the man at 25 m below the platform where he is first stopped by the rope.

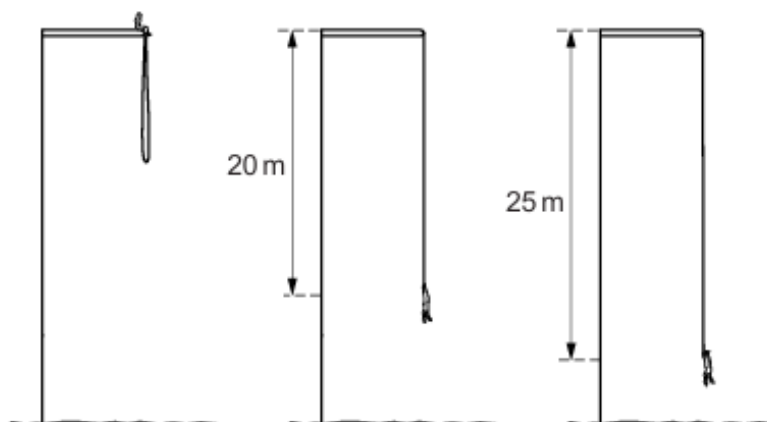


Fig. 3.1a

Fig. 3.1b

Fig. 3.1c

(a) As the man falls, his gravitational potential energy changes.

- (i)** The gravitational field strength is 10 N/kg. Calculate the change in his gravitational potential energy as he falls through 20 m.

change in potential energy =[2]

- (ii)** When he is 20 m below the platform, his kinetic energy is equal to the change in his gravitational potential energy calculated in **(i)**. Calculate his speed at this point.

speed =[3]

(b) State the energy changes that take place as he falls from 20 m to 25 m below the platform. Ignore the effect of air resistance.

.....

[3]

28. M/J/12/21/Q2

Fig. 2.1 shows a windsurfer, sail and board.

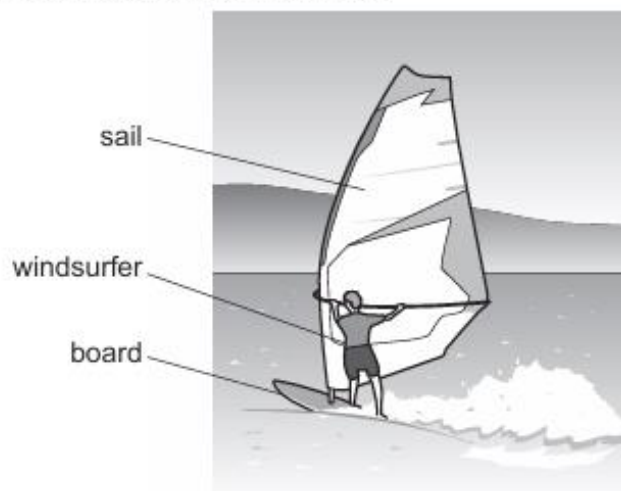


Fig. 2.1

The total mass of the windsurfer, sail and board is 90 kg. They travel at a constant speed of 5.0 m/s.

- (a) Calculate the total kinetic energy of the windsurfer, sail and board.

kinetic energy =[2]

- (b) The wind exerts a forward force on the sail.

- (i) Explain, in terms of the forces acting, why the board travels at a constant speed.

.....
.....
.....[2]

- (ii) The wind does work on the sail. Explain, in terms of the conservation of energy, why the total kinetic energy does not increase.

.....
.....[1]

29. M/J/11/21/Q3

Fig. 3.1 shows part of a small hydroelectric system that generates electricity from moving water.

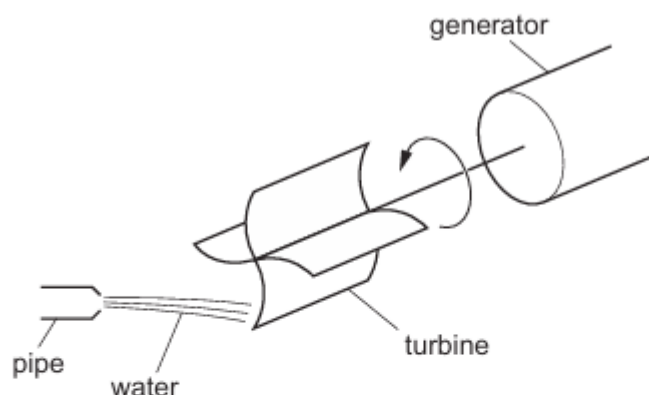


Fig. 3.1

- (a) Every minute, water with kinetic energy 14 000 J emerges from the pipe. The water turns a turbine that is connected to the generator.
Half of the kinetic energy of the water is given to the generator.

- (i) Calculate the power input to the generator.

power = [3]

- (ii) Suggest a reason why only some of the kinetic energy of the water is given to the generator.

.....
..... [1]

- (b) Hydroelectric systems use sources of renewable energy.

- (i) State what is meant by *renewable energy*.

.....
..... [1]

- (ii) State one source of **non**-renewable energy.

..... [1]

30. O/N/10/21/Q3

- (a) State the *principle of conservation of energy*.

.....
.....
..... [2]

- (b) Fig. 3.1 shows a miner using a head torch powered by a battery pack. The battery pack contains the cells that power the filament lamp in the head torch.

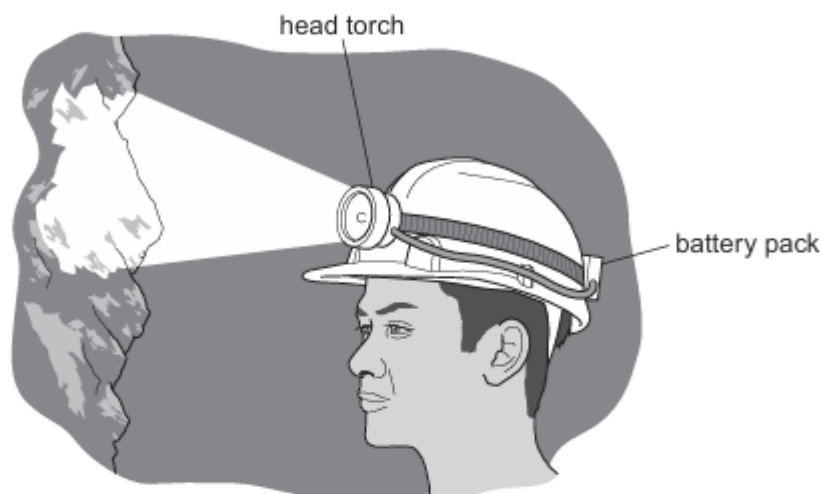


Fig. 3.1

- (i) State the main energy change that takes place in the cells.

.....
..... [2]

- (ii) State two different forms of energy that are produced in the filament lamp.

1.
2.
[2]

- (c) An identical battery pack powers a different head torch that uses light-emitting diodes (LEDs) instead of the filament lamp. The two torches are equally bright but the efficiency of the torch using the LEDs is much higher.

State how the amounts of the different forms of energy produced in the LEDs compare with the amounts produced in the filament lamp.

.....
.....
..... [2]

31. M/J/10/22/Q2

Fig. 2.1 shows a motor used to operate a lift (elevator). There is a man in the lift.

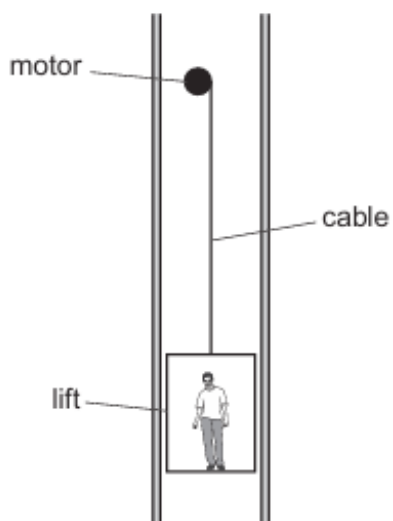


Fig. 2.1 (not to scale)

The input power to the motor is 6200 watts.

(a) Define the *watt*.

.....
 [2]

(b) The lift and the man have a total mass of 580 kg. The lift moves up a distance of 12 m in 15 s.

(i) Calculate the tension in the cable when the lift is moving at constant speed.

tension = [1]

(ii) Calculate the increase in potential energy of the lift and man.

potential energy = [2]

(iii) Calculate the efficiency of the motor.

efficiency = [2]

32. M/J/10/21/Q3

Fig. 3.1 shows a student rubbing her hands together.

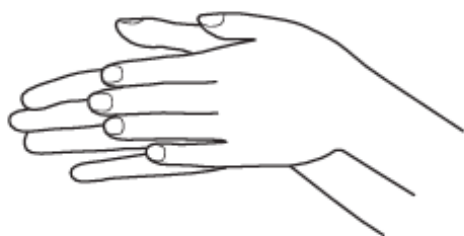


Fig. 3.1

- (a) State the main energy conversion that causes the hands to become warm.

.....
.....
.....[2]

- (b) State why the hands become even warmer if they are pressed harder together when rubbing.

.....
.....[1]

- (c) The average force used to slide one hand along the other is 1.2 N.
In each movement, one hand moves 0.080 m. The other hand remains stationary.

- (i) Calculate the number of movements needed for 2.0 J of work to be done.

number =[3]

- (ii) Each movement takes 0.20 s. Calculate the average power developed.

power =[2]

33. O/N/09/Q2

Fig. 2.1 shows a boat stationary in still water in a narrow tunnel.

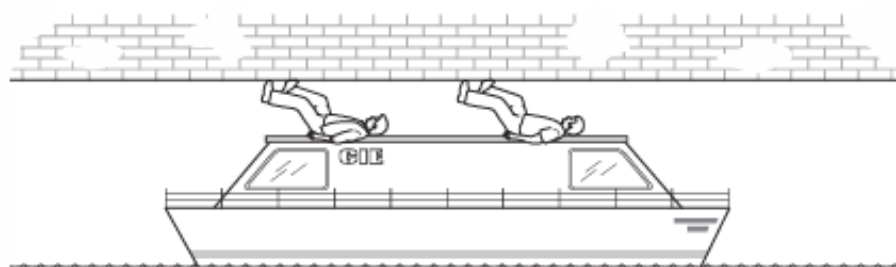


Fig. 2.1

To move the boat, two men lie on the top of it and push against the tunnel roof with their legs. They exert a total forwards force of 1680 N on the boat.

- (a) The men push the boat 50.0 m to the end of the tunnel. Calculate the work done on the boat by the men.

work done = [2]

- (b) Explain why the kinetic energy of the boat at the end of the tunnel is less than the work done on the boat by the men.

.....
.....
..... [2]

34. M/J/09/Q10

Fig. 10.1 shows the horizontal forces as a cyclist travels forwards.



Fig. 10.1

The cyclist produces the driving force that acts on the back wheel.
In this question, you may ignore any frictional force acting on the front wheel.

(a) The bicycle accelerates until a constant speed is reached.

(i) Describe how the size of the air resistance changes during this time.

.....
.....
..... [2]

(ii) Compare the sizes of the two horizontal forces when the bicycle is accelerating.

.....
..... [1]

(b) The total mass of the bicycle and the cyclist is 75 kg. At one instant, the speed of the bicycle is 4.0 m/s, the driving force is 30 N and the air resistance is 20 N.

Calculate

(i) the total kinetic energy of the bicycle and the cyclist,

.....
.....
.....
..... [3]

- (ii) the acceleration of the bicycle and the cyclist.

.....
.....
..... [2]

- (c) As the bicycle moves, energy is transmitted from the pedals to the back wheel.

Fig. 10.2 shows what happens to the energy input to the pedals.

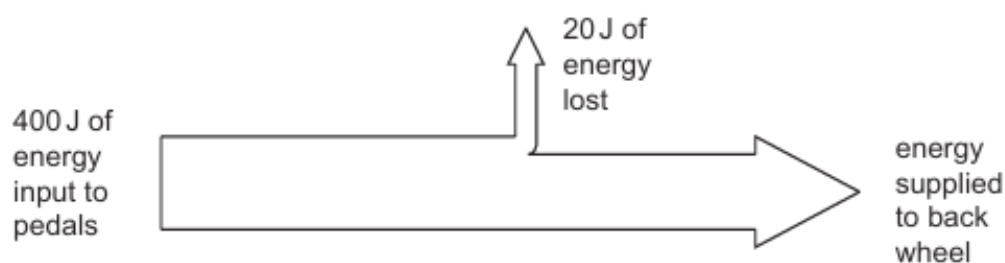


Fig. 10.2

- (i) As energy is transmitted to the back wheel, some is lost. Explain how this happens.

.....
.....
..... [2]

- (ii) Calculate the efficiency of the bicycle in transmitting energy from the pedals to the back wheel.

.....
.....
..... [2]

- (d) Some bicycles are made from low density materials. Explain why this is an advantage.

.....
.....
.....
..... [3]